MIL-P-81001(AS)
AMENDMENT-1
8 September 1970

MILITARY SPECIFICATION

POWER SUPPLY PP-2581(*)/A

This amendment forms a part of Military Specification MIL-P-81001(WP) dated 15 February 1965 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-P-81001(WP) is changed to MIL-P-81001(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Pages 1 and 2

2.1, under SPECIFICATIONS, Military: Delete

"MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists"

Page 1

2.1, under SPECIFICATIONS, Military: Add

"MIL-D-1000	Drawings, Engineering and Associated Lists"

Page 2

2.1, under STANDARDS, Military: Add

"MIL-STD-781	Reliability Tests: Exponential Distribution"

3.1, line 3: Delete and substitute

"(1) Parts and materials requirements shall conform to MIL-E-5400, (including electron tubes, transistors and diodes)."

FSC 5821

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Page 2 (Continued)

3.1.2: Delete entire paragraph.

Page 3

3.2.2, last line: Delete and substitute

"Input Power 115 ±1.0 V rms
400 ±2 cycles per second,
1-phase, and 27.5 ±0.5 V DC"

3.2.6.2: Delete and substitute

"3.2.6.2 Reliability in Mean Time To Failure (MTTF) - The equipment shall have a mean (operating) time to failure (MTTF) of 750 hours or greater when tested and accepted as outlined under the requirements of 4.4.3."

Page 4

3.3.1: Add new subparagraph

"3.3.1.2 High-Voltage Regulated Supply - Provides high voltage DC power to the missile.

(1) The output voltage shall be regulated at 175 plus or minus 15 volts DC over the load range of 5 to 175 milliamperes under any normal steady state condition of MIL-STD-704 input power as defined in 3.2.7.

(2) The open-circuit voltage shall not exceed 200 V DC under any electrical or environmental conditions. No damage shall result from open-circuit operation or from a 1-minute load of 280 milliamperes.

(3) The maximum ripple voltage shall not exceed 0.6 volt peak-to-peak under standard conditions and full load. Under any condition of voltage, frequency, load, and operating temperature the ripple shall not exceed 1.5 volts peak-to-peak. All ripple voltages consisting of frequencies other than the fundamental and second harmonic of the input AC supply shall be ignored.

(4) Under any transient of input voltage specified in 4.2.4.2.4, the output voltage shall not exceed 262 ±2 volts DC.

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Page 4 (Continued)

(5) These requirements shall apply over the specified temperature range.

(6) The 175 V DC power supply shall be capable of withstanding continuous short circuit loading without damage. "

Page 5

3.3.3 (1) and (2), first line: Delete "K-3" and substitute "K-303".

Page 6

3.3.5.2, first line: Delete "K-1" and substitute "K-301".

3.3.5.4, first line: Delete "K-5" and substitute "K-305".

3.3.5.5, first line: Delete "K-6" and substitute "K-306".

Page 7

4.2.4, second sentence: Delete and substitute

"Preproduction tests shall include environmental tests in accordance with 3.2.3.1, interference tests in accordance with MIL-I-6181, reliability index determination in accordance with MIL-R-22973(WEP), and the exceptions and modifications below. "

4.2.4.1, second sentence: Delete and substitute

"Resistance measurements shall be taken from J1 pin 18 to ground (missile signal output transformer). "

4.2.4.2.1, second line: Delete "J1" and substitute "J2".

4.2.4.2.4: Delete and substitute

"4.2.4.2.4 Transient Test - With the power supply operating with input power as defined in 4.2.4.2.2 Condition C and the power supply loaded as defined in 4.2.4.2.3 Condition A the AC input voltage shall be raised from 0 volts rms to 190 volts rms, maintained for a period of 100 (plus or minus 5) milliseconds and then dropped to test condition value. The rise and fall times shall not exceed 3 cycles. The output voltages shall not exceed allowable maximums. The above cycle shall

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Page 7 (Continued)

be repeated three times. Under the previous conditions of input power and loading, the input voltage shall be raised from 120 volts to 190 volts rms, maintained for a period of 100 (plus or minus 5) milliseconds and then dropped to 120 volts rms. The rise and fall times shall not exceed 3 cycles. The output voltages shall not exceed allowable maximums. This cycle shall be repeated three times. The six cycles, as detailed above, shall constitute the transient test."

Page 8

4.2.4.9, sentence beginning on line 5 at top of page 9: Delete and substitute

"This temperature cycle shall be repeated for a period of 200 hours and tested with the loads as indicated in 4.2.4.2.3.C."

Page 11

4.4.2.5: Delete and substitute

"4.4.2.5 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Power and Loads	Condition A of 4.2.4.2.2 and C of 4.2.4.2.3

During the 6-hour period of operation, simulated missile firings shall be conducted at 20-minute intervals. Should a failure occur, it should be repaired and the test started over. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from all undelivered equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours."

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Page 11 (Continued)

4.4.3: Delete and substitute

"4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted as required by MIL-STD-781. Equipment selected for Reliability Assurance Tests shall first have passed the individual tests."

Page 12

4.4.3.1: Delete entire paragraph.

4.4.3.2: Delete and substitute

"4.4.3.2 Test Level - The test level shall be E from Table I of MIL-STD-781, modified in accordance with Figure 1 as follows:

Cooling Cycle: With the power supply non-operating, time to stabilize at -65° F.

Heating Cycle: With the power supply operating in accordance with 4.2.4.2.3.C, the following procedure shall be adhered to:

TIME	ALTITUDE	TEMP. CHAMBER	INPUT POWER
Step 1 - 1/4 Hour	Sea Level	-65° F	105 V 420 cps
Step 2 - 1/4 Hour	80,000 Feet	-65° F	105 V 420 cps
Step 3 - Increase Chamber Temp. to +70° F (Approx. 1/4 Hour)	80,000 Feet	+70° F	105 V 420 cps
Step 4 - Continue to increase Chamber Temp. to 131° F	Sea Level	+131° F	122 V 380 cps
Step 5 - Maintain Chamber Temp. for such period that the overall heating and cooling cycle time will be 6 hours."	Sea Level	+131° F	122 V 380 cps

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4.4.3.3: Delete and substitute

"4.4.3.3 Accept-Reject Criteria - Test Plan V of MIL-STD-781 shall be used to determine the accept-reject criteria for the Reliability Qualification Test, and Test Plan VI shall be used for the Reliability Sampling Test."

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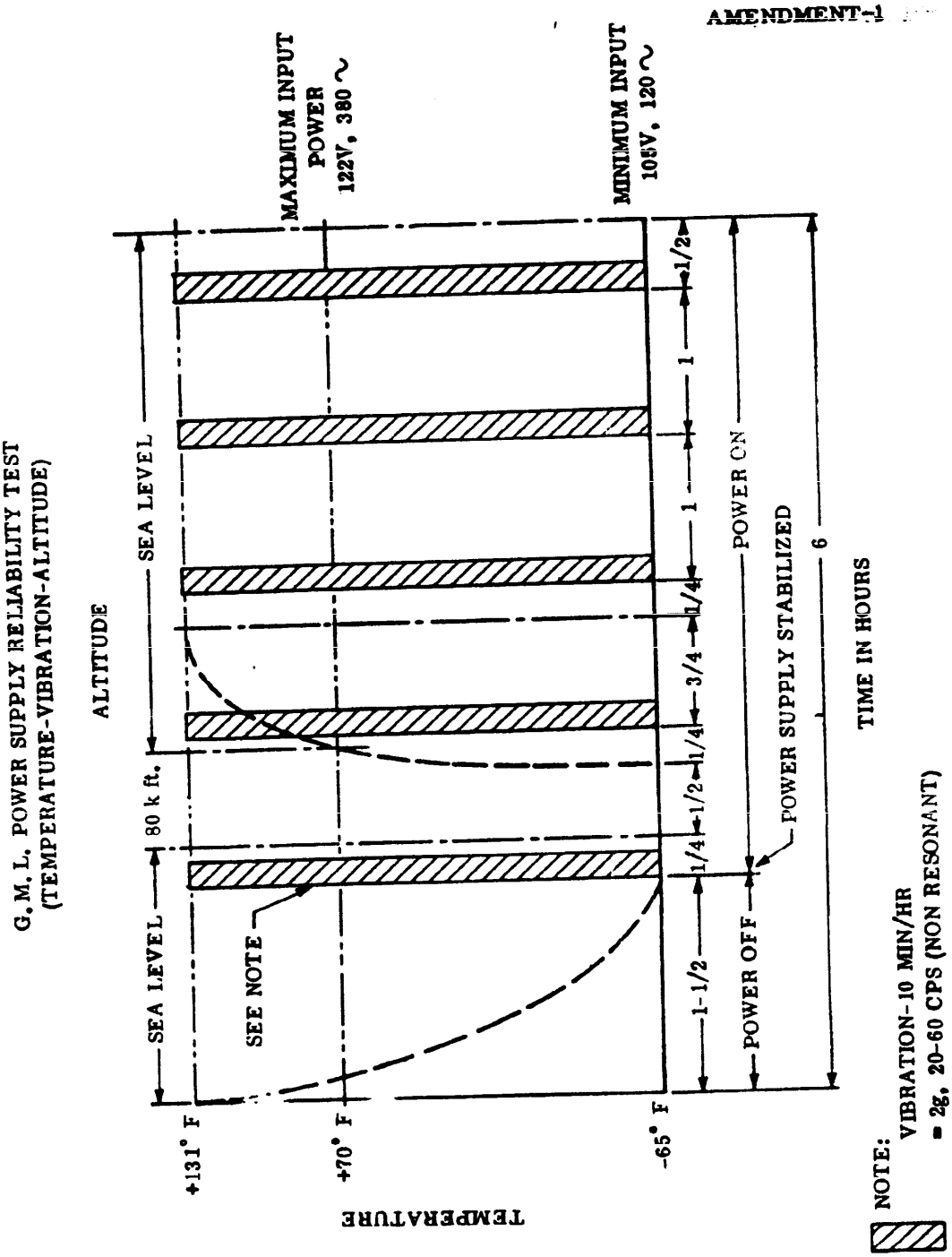


Figure 1. Test levels

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